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# 13

# Parametrics and the Limits of Knowledge

In chapters 8–12, I explored how scientists use models to process, interpret, and reconstruct observations, to *make data global*. Models are how we know what we know about the history of climate. They are also why we can keep learning more about the history of climate, without ever securing a single definitive version of that history once and for all. The past shimmers. What about the future?

If we need models to make sense of data about the past, we need them all the more to predict whether, and how, the climate will change. Chapter 7 described how climate models work. Recall the main difference between general circulation models of weather and climate: Climate GCMs don't start from observations. Instead, they simulate their own climates. Then we compare their simulated climates with the real one—or, rather, with the data about the climate that we have (re) constructed, *made global*, in all the ways the last five chapters have described.

Climate models don't use data, I said—and this is true, in the sense that they don't ingest observations of the real atmosphere and use them to correct their simulations, as weather forecast models or data assimilation systems do. Yet climate models use data in many other, very important ways. This chapter is about a fuzzy boundary between data, theory, and algorithms, a place deep within the practical, everyday work of general circulation modeling where modelers combine measured quantities with code to calculate the effects of processes too small, too complex, or too poorly understood to be modeled directly. These are the "semi-empirical" parameter schemes by which modelers handle sub-grid-scale processes, known as the "model physics." (See chapter 7.) Later in the chapter, I will talk about how modelers "validate," or "verify," or "evaluate" climate models, a perilous choice among seemingly similar terms that amounts to the $100 trillion question for climate modeling: Can we trust model predictions of climate change?

'Parameter' is an interesting word, meaning literally a "para-metric" or "para-measure." The prefix 'para', from Greek, can mean many things, including beside, near, along with, beyond, resembling, faulty, undesirable, assistant, and auxiliary. According to a glossary maintained by the American Meteorological Society, 'parameter' refers specifically to "any quantity of a problem that is not an independent variable. More specifically, the term is often used to distinguish, from dependent variables, *quantities that may be more or less arbitrarily assigned values for purposes of the problem at hand*" (emphasis added). So a parameter is a kind of proxy—a stand-in for something that cannot be modeled directly but can still be estimated, or at least guessed.

Parameterization illustrates the interaction of computational friction with the limits of human knowledge. In an ideal climate model, the only fixed conditions would be the distribution and the altitude of continental surfaces. Virtually all other variables—sea-surface temperature, land-surface albedo (reflectance), cloud formation, etc.—would be generated internally by the model itself from

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the lower-level physical properties of air, water, and other basic elements of the climate system. Instead, *most* physical processes operating in the atmosphere require some degree of parameterization; these parameterized processes are known as the "model physics." Generating these parameters (or "parameterizations," as scientists somewhat confusingly call them) is a large part of any climate modeler's work.[1]

## Types of Parameterization

Parameters may be fixed quantities, such as coefficients, or they may be mathematical functions containing both coefficients and dependent variables. The simplest fixed parameters are specified directly from empirical measurements. These include such quantities as solar radiation, the size and location of continents, the gravitational constant, and the concentration of carbon dioxide in the atmosphere.

Much more commonly, parameters represent a variable physical process rather than a fixed quantity. An easy way to think about this is to imagine modeling rainfall. In principle, you could try to model individual raindrops, but it's hard to see how that will help you if you're interested in rain over an area the size of Ohio (about the size of today's climate model grid cells). You just want to know how much water fell on the whole state, and when, so your model of rainfall need not include individual drops.

A major parameterization in all climate models is radiative transfer. The atmosphere contains both gases ($CO_2$, methane, nitrogen, ozone, oxygen, water vapor, etc.) and solids (particulate aerosols, ice clouds, etc.). Each one of these materials absorbs solar energy at particular frequencies. Each also emits radiation at other frequencies. Those emissions are then absorbed and re-radiated by other gases and solids. These radiative transfers play a huge role in governing the atmosphere's temperature. Thus, models must somehow estimate how much radiation the atmosphere in a given grid box absorbs, reflects, and transmits, at every level and horizontal location. This is the problem discussed at the end of chapter 7, which Callendar, Plass, Möller, Manabe, Rasool, and Schneider all tried to tackle, each adding a new level of understanding of the problem's complexity. "Line-by-line models," which combine databases of spectrographic measurements for the various gases with physical models, can carry out this summing.[2] Today's best line-by-line models achieve error margins better than 0.5 percent. Yet the computational cost of line-by-line models precludes using them directly in climate GCMs. Therefore, 1980s GCMs used fixed parameters derived from line-by-line model results to approximate actual radiation absorption in a given grid box on the basis of its specific mixture of gases, the angle of the sun's rays at that location, and other factors. Today's GCMs go further, incorporating "fast radiative transfer models" that can calculate radiative interactions among various gases and radiative interactions with aerosols, clouds, ice clouds, and other features of the atmosphere. Modelers have systematically compared these fast radiative transfer codes (i.e., parameterizations) against the results of line-by-line models, over time bringing them ever closer together. At this writing, the best fast radiative transfer models contained in climate GCMs track the results of line-by-line models to within about 10 percent.[3] The nesting of complex subprocess models within larger GCMs exemplifies the intense interplay among models that has become a hallmark of computational science.

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Let's take another example: cloud physics. Clouds play crucial roles in climate. Convection processes responsible for towering cumulus clouds are a primary mechanism of vertical heat transport in the atmosphere. Clouds also strongly affect surface temperature; they can raise it or lower it, depending on conditions. For example, clouds can form a heat-trapping "ceiling" at night, yet their white tops reflect incoming solar energy back into space during the day, cooling the surface. Clouds' climatic effects depend on many factors, including their altitude, density, and color. They interact with aerosols, which can provide nuclei around which cloud droplets form, and which can also affect clouds' color, changing their reflective characteristics.

Because clouds affect climate so greatly, and because they are still so poorly understood, cloud parameterization remains among the most controversial areas of climate modeling. Clouds develop from "complicated interactions between large-scale circulations, moist convective transport, small-scale turbulent mixing, radiation, and microphysical processes," and they remain notoriously difficult to study empirically.[4] Many kinds of clouds form on scales of a kilometer or less, two orders of magnitude below climate GCM resolutions. Therefore, rather than represent cloud formation directly—in terms of convection columns, condensation nuclei, and their other physical causes—a GCM might calculate the amount of cloud cover within a grid box as some function of the independent variables temperature and humidity. To deal with the issue of scale, climate GCMs typically calculate a "cloud fraction" (the percentage of each grid box covered by clouds). This raises the question of what to do when clouds appear at two or more levels in the same atmospheric column. If a grid box on one level is 50 percent cloudy and another grid box two levels higher is also 50 percent cloudy, how much of the sky is cloudy at ground level: all of it? 60 percent? 75 percent? GCMs must be provided with functions that determine how much overlap to specify; this can depend on other variables.[5]

Some cloud parameterization schemes in the earliest GCMs resulted in cloud "blinking," an oscillation between the presence and absence of cloud cover in a given grid box at each time step when certain variables lay just at the critical threshold. (Real clouds do not, of course, behave this way.) Cloud parameterizations remain problematic even in relatively recent models. The Rosinski-Williamson study of round-off error, described in chapter 7, discovered that such errors affected cloud parameterization in NCAR's CCM-2 far more than they affected the model's dynamical core, owing to interaction of the equation specifying cloud behavior with the fixed maximum value specified for relative humidity in the parameterization.[6]

The list of model parameters and their interactions is very long, and many of these are just as problematic as those described above. Other phenomena not well captured by parameterization include the planetary boundary layer (the turbulent layer of air nearest the Earth's surface) and such characteristics of land surfaces as roughness, elevation, and heat retention. For example, climate models of the early 1990s, operating at a resolution of 500 km, represented the entire region between the Sierra Nevada and the Rocky Mountains as a single plateau of uniform elevation. Even today's 200-km GCMs can scarcely separate California's Central Valley from the mountain ranges on either side of it. Much less can they resolve the 50×120 km bowl of the San Francisco Bay, which also is surrounded by mountains.

Modelers develop parameterizations by reviewing observational data and the meteorological literature. They identify the range of observed values and try to find relationships between small-scale processes and the large-scale independent variables in their models. When they succeed in

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finding such relations, they call the resulting parameters "physically based." Often, however, they do not find direct links to large-scale physical variables. In this quite common case, modelers invent *ad hoc* schemes that provide the models with the necessary connections.

An example of an ad hoc parameter is "flux adjustment" in coupled atmosphere-ocean circulation models (AOGCMs). The interface between the atmospheric model and the ocean model must represent exchanges of heat, momentum (wind and surface resistance), and water (precipitation, evaporation) between the atmosphere and the ocean. These fluxes—flows of energy and matter between atmosphere and ocean—are very difficult to measure empirically. Yet they profoundly affect model behavior. From the 1970s until the mid 1990s virtually all AOGCMs included *ad hoc* parameters, known as "flux adjustments," that artificially altered the modeled fluxes to keep them in line with observed values.[7] Modelers spoke of flux adjustments as "non-physical" parameterizations—i.e., ones not based on physical theory—but also sometimes characterized them as "empirically determined."[8] In 1997, version 2 of NCAR's Community Climate System Model became the first AOGCM to function realistically and stably during long runs (centuries) without flux adjustment.[9] Since then, many other major climate models have also been weaned from flux adjustment, but a small minority of models included in the 2007 IPCC assessment still used the technique.[10]

Any given GCM's model physics contains hundreds or even thousands of parameterizations. The model's overall outputs depend on the interactions among all of these parameterizations, as well as on how they interact with the model's dynamical core (equations of motion). The extreme complexity created by all these interactions can make it difficult to determine exactly why models do what they do. An entire subfield—climate model diagnosis—works out ways to isolate the origin of particular problems to specific parameterizations and their interactions.

## Tuning

Another important part of modelers' work is known as "tuning" the parameters. "Tuning" means adjusting the values of coefficients and even, sometimes, reconstructing equations in order to produce a better overall model result. "Better" may mean that the result agrees more closely with observations, or that it corresponds more closely to the modeler's expert judgment about what one modeler I interviewed called the "physical plausibility" of the change. In some cases, parameters are well constrained by observational data. In such cases, you know the range of values found in the real world precisely, so if your parameterization takes you outside that range, you know that you must fix it. In other cases, empirical data don't constrain the range so much. A variety of values and ranges may all be plausible within the limits of existing knowledge, so you have to make some choices, guided mainly by scientific intuition—and by how your choices affect model behavior. Such parameterizations, which include clouds and aerosol effects, are said to be "highly tunable." Since many parameters interact strongly with others, changing one parameterization may push the behavior of others outside an acceptable range, requiring further tuning.

Since the mid 1990s, modelers have increasingly focused attention on parameterization and tuning of aerosol effects in GCMs. Aerosols are microscopic airborne particles, such as dust, soot, and sea salt. Some are blown into the air. Others are formed in the atmosphere by interactions among gases; their origins are both natural (dust, salt) and human (automobile exhaust, coal soot).

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Aerosols influence climate in numerous ways, most importantly by altering cloud albedo (reflectance) and by directly reflecting or absorbing heat. As with clouds, much of their behavior derives from very small-scale interactions, and they are notoriously difficult to study *in situ*. High-resolution aerosol models now exist that can treat these processes explicitly on scales up to a few kilometers, but their computational demands are too great for use in GCMs. Hence parameterizing aerosols in regional and global models remains a work in progress.

Though each particular aerosol's properties are different, overall aerosols seem to exert a cooling influence on climate (figure 13.1). Yet the range of uncertainty regarding aerosols' climatic effects is large. At one extreme of that range, aerosols could exert an influence nearly as large as those of all anthropogenic greenhouse gases together, *but opposite in sign*.[11] Climate scientists say that aerosols "mask" the warming effects of greenhouse gases, because their lifetime in the atmosphere is short (days to months). If aerosol production were to fall dramatically—for example, because coal-fired power plants adopted exhaust scrubbing systems—aerosol levels in the atmosphere would drop quickly and the warming effects of greenhouse gases would assert rapidly themselves. Further, as the atmosphere warms, it becomes more humid. This tends to increase aerosols' dropout rate, reducing their lifetime in the atmosphere enough to decrease aerosol levels even without emissions reductions.

This strong sensitivity of the climate system means that getting aerosol parameters right matters greatly to modelers' ability to project future climate change. The crude aerosol parameterizations used in models for the earliest reports of the Intergovernmental Panel on Climate Change resulted in an untenably high projected warming. When Mount Pinatubo erupted in 1991, ejecting a huge dust cloud that reduced global temperatures by about 0.5°C for a number of months, attention to aerosols and their parameterization increased dramatically. Yet our understanding of aerosol chemistry and physics remains much more limited than our understanding of greenhouse gases and other important climate forcings, as reflected in figure 13.1's long error bars for aerosol effects. Because of this, Jeff Kiehl and others have recently raised concerns about the fact that the overall outputs of GCMs seem to agree more closely than their aerosol parameterizations should permit. The (unconfirmed) suspicion is that modelers are adjusting these "highly tunable" parameters, perhaps more than they can justify from empirical knowledge, in order to keep overall model outputs within a plausible range.[12] Nothing is likely to improve current climate models more than better, less highly tuned parameterizations of aerosol effects.

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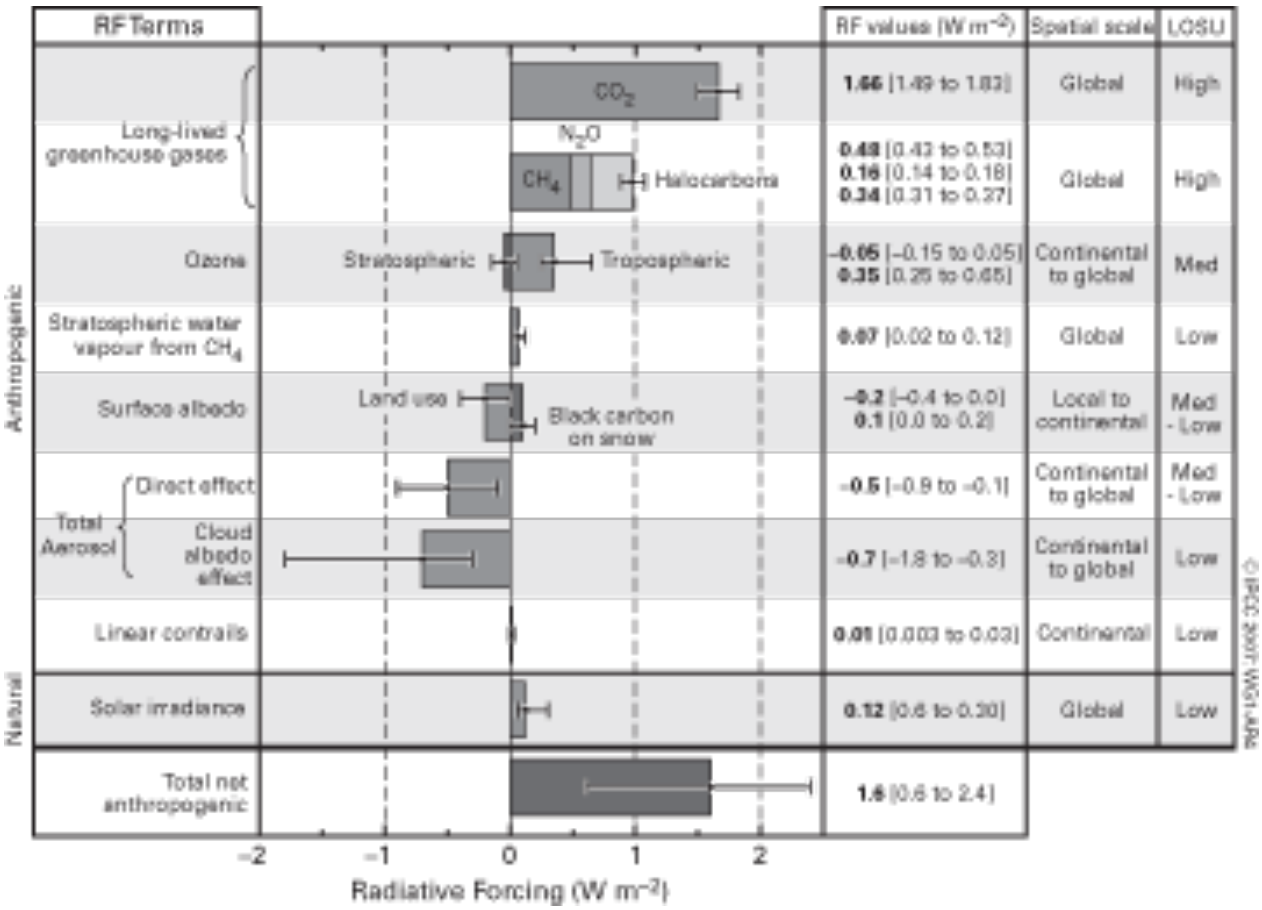


**Figure 13.1**
The major global radiative forcing factors, with uncertainties indicated by black bars. At the extremes of their uncertainty range, the combined negative forcing from direct and indirect aerosol effects could be at least as large as the positive forcing caused by long-lived greenhouse gases. LOSU (rightmost column) represents the IPCC-assessed "level of scientific understanding." *Source*: *Climate Change 2007: The Physical Science Basis* (Cambridge University Press, 2007). Image courtesy of IPCC.

## Living in the Semi-Empirical World

Parameterization and tuning remain, in effect, scientific art forms whose connection to physical theory and observational data varies widely from case to case. As Myanna Lahsen, Simon Shackley, and other ethnographers of climate modeling have observed, modelers can become so deeply involved in adding parameters and tweaking them to adjust model output that they lose track of the physical justifications for these practices.[13] One of Lahsen's interviewees, an empirical meteorologist, told her:

> Building a model is a full-time job and it takes a lot of expertise just to do that, and I think that sometimes it's easy to lose track of what the model is being built for. It's not enough simply to build a sophisticated model that includes a lot of physics, physical parameterization, etc., and to integrate the model in time and to take just sort of a general view of it, saying, "yeah it looks like it's simulating something that appears to be like the real atmosphere." . . . It's just as important to develop a model but, at the same time, to fully realize the limitations of the model and to always take a very critical look at the capabilities of the model and what kind of questions one may be able to answer and pursue using the model. The climate system is so complex, with all of the feedback loops, that it's very difficult to actually model all of those loops.[14]

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Or, as one climate modeler told me: "Sure, all the time you find things that you realize are ambiguous or at least arguable, and you arbitrarily change them. I've actually put in arguable things, and you do that all the time. You just can't afford to model all processes at the level of detail where there'd be no argument. So you have to parameterize, and lump in the whole result as a crude parameter." Another Lahsen interviewee described the psychological effect of doing model-based research as making it hard to differentiate the model from reality:

> You spend a lot of time working on something, and you are really trying to do the best job you can of simulating what happens in the real world. It is easy to get caught up in it; you start to believe that what happens in your model must be what happens in the real world. And often that is not true. . . . The danger is that you begin to lose some objectivity on the response of the model [and] begin to believe that the model really works like the real world. . . .[15]

Another ethnographer, Simon Shackley, described two "epistemic lifestyles" in climate modeling. In his terms, *climate seers* use models to "understand and explore the climate system, with particular emphasis on its sensitivity to changing variables and processes," seeing the models as tools for this purpose. Meanwhile, *climate model constructors* see models as an end in themselves. They seek to "capture the full complexity of the climate system [in models] which can then be used for various applications." Climate model constructors are more likely to focus on increased "realism," an adjective referring not to accuracy but to the inclusion in the model of all physical processes that influence the climate. Climate seers, by contrast, tend to focus on modeling the most fundamental and best understood processes, and to use a variety of different models, including simpler zero-, one-, and two-dimensional models. "For the seer," Shackley writes, "*which model is 'state-of-the-art' depends on the model's intended application*. . . . For the model constructor, by contrast, *a single state-of-the-art model exists irrespective of its application*. It is defined as one whose control is close to the observed climate, which uses the most up-to-date observations as inputs and for verification, and which contains the most detailed, most physically realistic parameterizations."[16] This distinction, Shackley points out, does not necessarily divide good modeling practices from bad ones. Rather, it represents two legitimate strategies aimed at producing different kinds of knowledge. Climate seers may be more prone to tune their models according to intuitive judgments of how model results can be "improved," but they may also insist on simpler parameterizations whose physical basis and effects on model output can be more readily understood than the more "realistic" but more complex parameterizations preferred by model constructors. By contrast, model constructors' efforts at realism attempt to capture the full complexity of the atmosphere, seeking advances in knowledge from the potentially unexpected results of adding more processes to the models. The model constructors' approach also has a political dimension, since those who challenge GCM results often argue that they do not account for the effects of some unincluded process, such as cosmic rays.[17] Adding more processes reduces modelers' vulnerability to this line of attack, though at the same time it increases the opportunities to question the accuracy of parameterizations.

Controversies about tuning rage both inside and outside the climate modeling community. The philosopher-physicist Arthur Petersen notes that "simulationists hold divergent views on the norm of not adding *ad hoc* corrections to models."[18] Some accept these corrections as necessary; others view them almost as morally suspect and seek to eliminate them. David Randall and Bruce Wielicki argue that tuning "artificially prevents a model from producing a bad result." Noting that some modelers refer to tuning as "calibration"—exploiting that term's positive connotations—

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Randall and Wielicki write: “Tuning is bad empiricism. Calibration is bad empiricism with a bag over its head.” Yet Randall and Wielicki also acknowledge that, in the case of physical processes that are known to be important but are not well understood, there may be no choice.[19]

In general, modelers view tuning as a necessary evil. Most try to observe certain constraints. Tuning should not, for example, bring the tuned variable outside its known range of observed behavior. In addition, models should be evaluated on observational data for which they have not been explicitly tuned. In other words, tuning lower-level variables may be justified if it produces better higher-level behavior, for example in reproducing multi-dimensional phenomena such as the North Atlantic oscillation or regional monsoons. Ultimately it may be possible to use techniques borrowed from data assimilation to tune models automatically, reducing the potential for subjective effects to influence the process.[20]

## Models vs. Data: Validation, Verification, or Evaluation?

In the 1960s and the 1970s, climate models were still too primitive for anyone to worry about getting the details right. Furthermore, as we saw in chapter 7, data on the full global circulation (especially the vertical dimension) remained sketchy until the 1978–79 Global Weather Experiment. In those days, you couldn’t be completely sure which side problems were coming from. How much of the disagreement between your model and the data was due to problems in the model, and how much to noise in the data?

In the 1980s, as both models and data improved and climate change became a high-stakes political issue, modelers began to look for more than ballpark resemblances. They now wanted to show that their models could capture—reliably and with some degree of precision—the real climate’s behavior. Until the end of that decade, there were essentially two ways to do this. The simplest was to set your model parameter values to present-day conditions, run the simulation for 20–100 years to get a climate, then compare your results with statistics for the actual circulation. The other way was to set your parameters to a *different* set of conditions, such as those of the last ice age or the hot Cretaceous. How would your model respond to changing the distributions of continents, the Earth’s orbit, solar input, carbon dioxide concentrations, and other factors characteristic of the distant past? If your model’s results reasonably resembled paleontological estimates of past climate conditions, that could boost your confidence in its skill. If setting your model’s parameters to present-day conditions also produced a reasonable result, this made it more likely that your simulation was accurate. From there, you could argue that setting its parameters to some future state—such as carbon dioxide doubling—could give you a reasonably accurate picture of how the climate would respond.

As they gradually became available in the early 1980s, data from the First GARP Global Experiment provided a much more detailed and reliable data image of the global circulation against which to compare GCM simulations. The FGGE data assimilation (see chapter 9) produced three-dimensional gridded data sets from which circulation statistics for the FGGE year could readily be derived. Modelers soon began to use these statistics as checks on the realism of GCM outputs. Owing to the short length of the record, however, *these were not climate data*. You needed a longer record to get a true climatic picture. Meanwhile, the Climatic Research Unit, the Goddard Institute for Space Studies, and other groups discussed in chapter 11 were assembling an increasingly detailed and widely trusted set of climate data for the surface.

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The increasingly precise data emboldened modelers to talk, by the end of the 1980s, about "verifying" or "validating" climate models.[21] A "verified" or "validated" model would be one that reliably reproduced observed climate. In the political context of that period, these terms soon became hotly contested. In 1994, Naomi Oreskes, Kristin Shrader-Frechette, and Kenneth Belitz argued in the pages of *Science* that talk of "verification" or "validation" of models was bad epistemology.[22] The word 'verification', they wrote, normally implies definitive proof. But models, Oreskes et al. argued, are essentially intricate inductive arguments. Since no inductive proposition can be proved with perfect certainty, models can never be verified in this strict sense. The fact that a model agrees—even perfectly—with observations does not guarantee that all of the principles it embodies are true. Furthermore, as was discussed in chapter 10, the logic of simulation modeling does not require, or even permit, definitive proof. For example, parameterizations by definition do not derive from exact physical principles; no one expects them to prove perfectly accurate.

Oreskes et al. saw "validation" as a somewhat less stringent standard. Strictly defined, it refers to demonstrated internal consistency and an absence of detectable flaws. On this reading of the term, a model might be "valid" without being an entirely accurate or complete explanation. Nevertheless, Oreskes et al. pointed out, scientists commonly used 'validation' and 'verification' as synonyms. If scientists did not see a difference between them, neither would policymakers or ordinary citizens.

Oreskes et al. argued that models can at best be "confirmed." This implies only that model results agree with observations. This agreement, by itself, tells us nothing about whether the model reached its results for the right reasons. Thus, "confirming" a model does no more than make it a viable candidate for a true explanation. In other words, confirmation raises the likelihood that the model embodies true principles, but does not make this certain. This view is consistent with Popper's well-known doctrine of falsificationism, which holds that scientific hypotheses (or models) can be proved false by observations, but cannot be proved true.[23]

This critique had a considerable impact on the Intergovernmental Panel on Climate Change, which dropped the words 'validation' and 'verification' altogether. (However, those words still appear regularly in the scientific literature reviewed in IPCC assessments.) The agency substituted 'evaluation', defined as an assessment of "the degree of correspondence between models and the real world they represent"[24]: a claim of relevance and trustworthiness, rather than truth.

By the late 1980s, increased computer power permitted the first efforts to simulate the actual climate of the twentieth century. This marked a major change in the role of GCMs. Previously, most studies simply compared a control run under pre-industrial conditions with a second run under $CO_2$ doubling. The product was a calculation of the "climate sensitivity" (a phrase which invariably refers to $CO_2$ doubling), the figure shown in tables 7.3 and 14.1. But in the real atmosphere, of course, $CO_2$ does not double overnight, and many other climate-related events occur at various times. Simulating the year-by-year course of twentieth-century climate would demand "transient forcing" runs, with $CO_2$ levels specified at their actual time-varying amounts and, perhaps, with major volcanic eruptions and other consequential events included in the appropriate years. Since climate is an average state, no one expected such transient runs to match the course of change exactly, but averaged over periods of 5–10 years or more a good simulation should roughly track the basic trends shown in figure 11.5. The first such transient runs were attempted, with some success, in the second half of the 1980s. (See figure 14.3.)

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## Model Intercomparisons as Standardized Gateways

After the establishment of the Intergovernmental Panel on Climate Change in 1988, modelers initiated a number of exercises in model evaluation. The most important of these exercises—and one that became crucial to integrating the climate knowledge infrastructure—was model intercomparison. The Atmospheric Model Intercomparison Project, mentioned in chapter 7, was among the first of these projects. It was established at the Lawrence Livermore International Laboratory in 1989, when the US nuclear weapons laboratories began seeking non-military uses for their supercomputer facilities and modeling expertise. The AMIP strategy required each modeling group to run its model using a specific set of "boundary conditions," or parameters (specifically, monthly average sea surface temperature and sea ice distribution for the years 1979–1988). All model runs had to provide specific output variables in a standard format. Whereas previously each modeling group had chosen boundary conditions and output variables for itself, these common specifications created a basis for intercomparing model performance and diagnosing the sources of differences in model behavior.

After the Atmospheric Model Intercomparison Project, intercomparison evolved into a series of more elaborate projects. Many of these participated in what became a regular cycle of model evaluation linked to the Intergovernmental Panel on Climate Change assessment process. Today, most intercomparison projects are organized under the umbrella Program for Climate Model Diagnosis and Intercomparison.[25] Some 40 different projects evaluate different aspects of all types of climate models. The current phase of the Coupled Model Intercomparison Project (CMIP), for example, compares model simulations of the following:

pre-industrial control run

present-day control run

climate of the twentieth century experiment (realistic greenhouse gas increases)

1 percent per year $CO_2$ increase to doubling and quadrupling

stabilization at 550 ppm $CO_2$

stabilization at 720 ppm $CO_2$

and numerous others, prescribing well over 100 output fields the models should generate to permit direct comparison.

In terms of the theory of infrastructure development developed in chapter 1, intercomparisons function as standardized gateways linking the various models and modeling groups. By permitting regular, direct, and meaningful comparisons of the models with one another and with standardized data sets, they have helped to transform climate modeling from a craft activity of individual laboratories into a more modular and standardized collective activity involving virtually all of the world's climate modeling groups; in theoretical terms, they linked a set of isolated systems and created a network. Technological change, of course, facilitated this transformation. With the rise of the Internet during the 1980s, modelers could exchange data far more easily.[26] The broad international participation and frequent interactions that characterize model intercomparisons today would not have been possible under previous information regimes. These projects can thus be seen as early versions of what would today be called "cyberinfrastructure" or "e-science."

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How do current GCM results compare with each other? How well do they track the twentieth century climate record? What do they tell us about the relative roles of natural and anthropogenic factors? Figure 13.2 provides a glimpse of all these comparisons simultaneously. In both top and bottom charts, the thick black line represents the Climatic Research Unit surface temperature data set discussed in chapter 11. The year-by-year outputs from numerous simulations fluctuate, as expected, around the actual climate record. (No climate model would *or should* reproduce the record exactly, due to climate's natural variability.) The average trend from all the simulations together is also shown.

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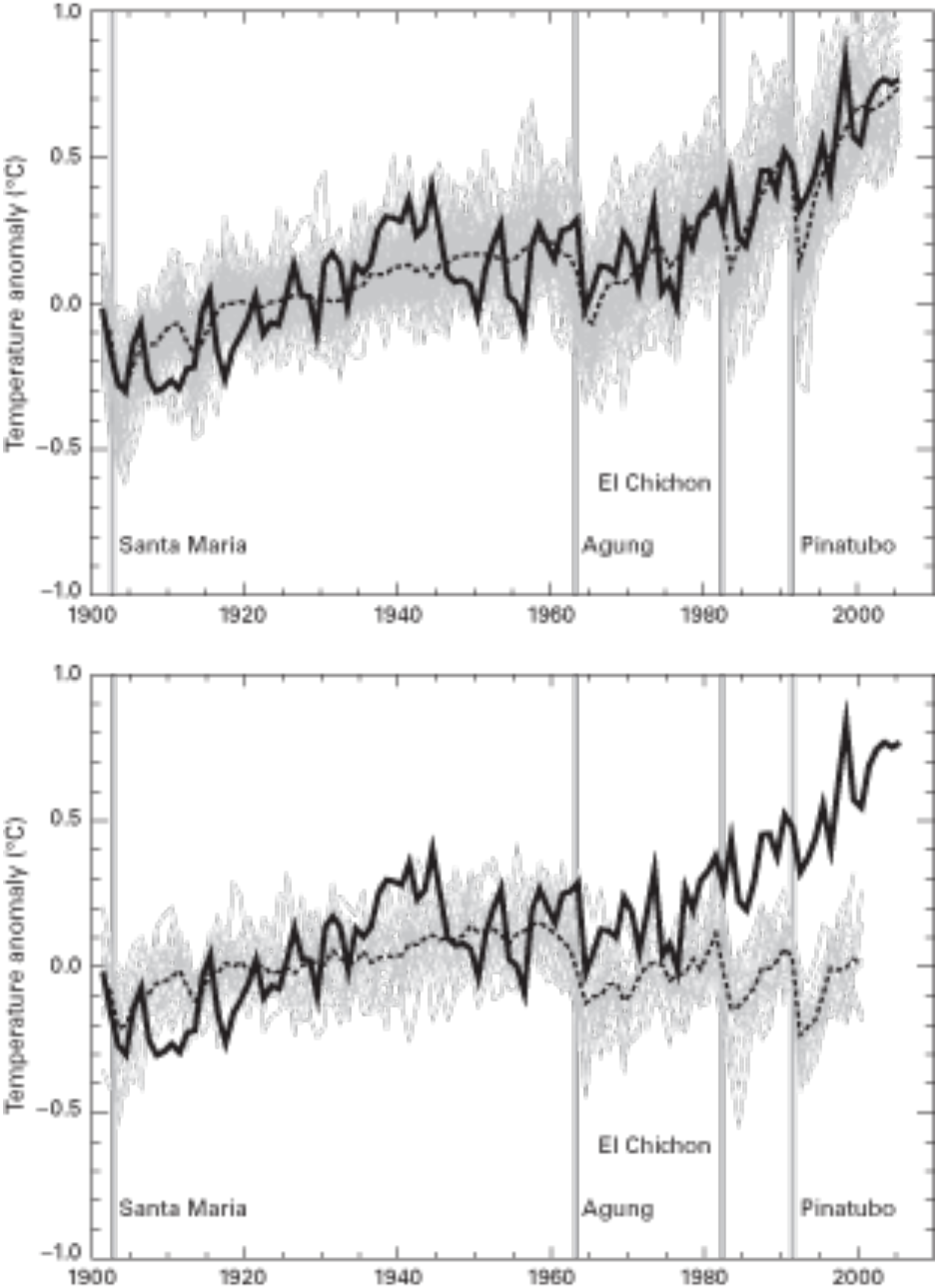


**Figure 13.2**
AOGCM simulations (thin gray lines) of twentieth-century global surface temperature anomaly with (top) and without (bottom) anthropogenic forcing. Black line in both graphs represents observations (HadCRUT3). Dashed lines represent the trend across all model simulations. Vertical gray lines mark major volcanic eruptions. *Source*: *Climate Change 2007: The Physical Science Basis* (Cambridge University Press, 2007). Image courtesy of IPCC. Colors replaced to allow non-color reproduction.

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Figure 13.2 shows that when both natural and anthropogenic forcings are included in the models, their average trend tracks the twentieth-century record reasonably well. However, when anthropogenic forcings (greenhouse gases, aerosols, and other human activities that affect the atmosphere) are removed, the model trend begins to fall below the observed one around 1960, departing from it altogether by 1980. This is true not just of the models' average; it is true of every individual model run too. *Not one* of the models compared here could reproduce the most recent period of warming without including human activities.

Figure 13.2 shows global averages, but since 2000 model resolution has increased to the point that modelers can calculate similar figures for climate on a continental scale. Figure 13.3 shows the same comparison as in figure 13.2, broken down by continent and also by land surface and ocean components. In all cases, model-calculated trends track observations (black trend lines) so long as anthropogenic factors are included, but diverge substantially by the last decades of the twentieth century if only natural forcings are used.

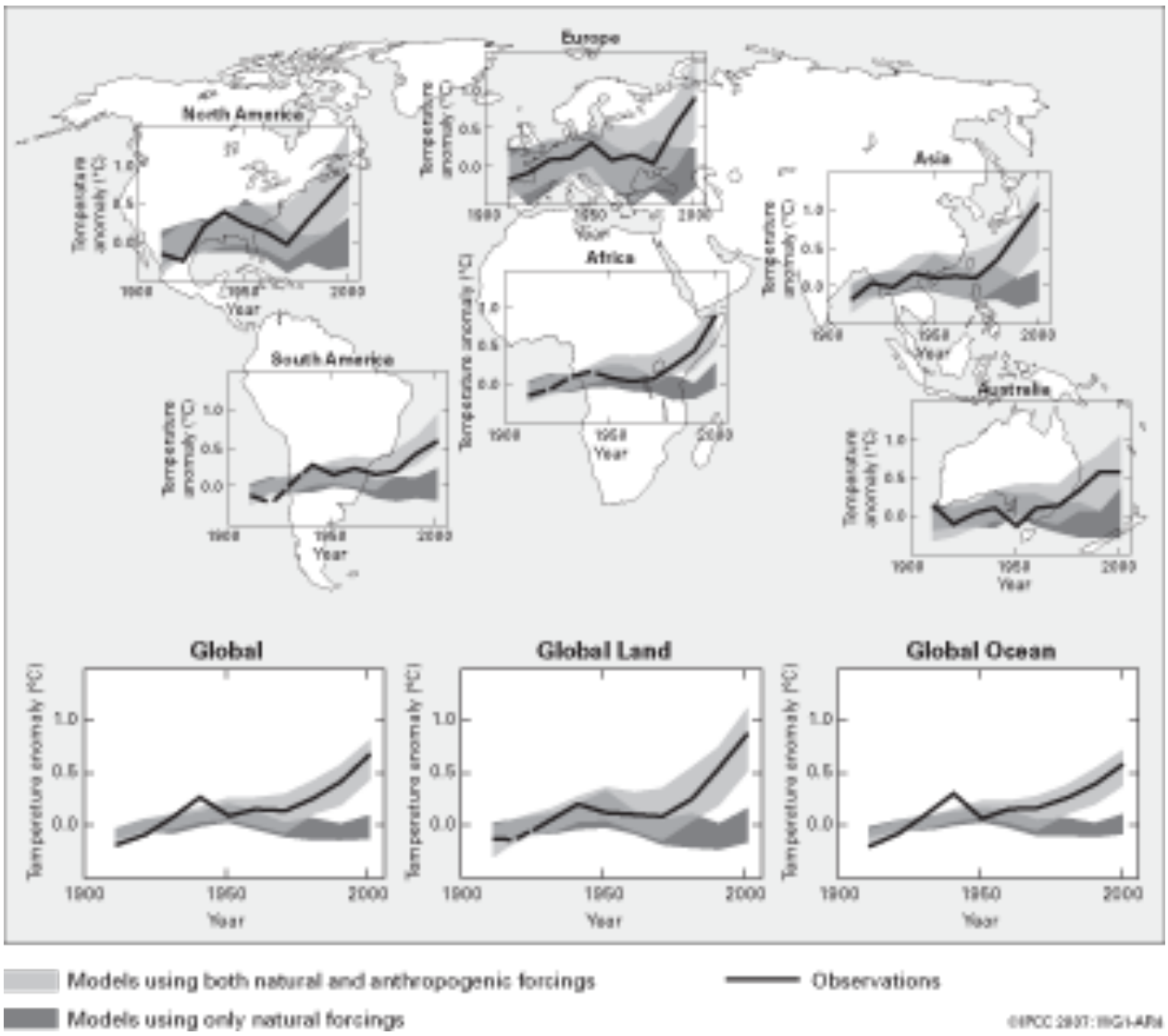


**Figure 13.3**
Original legend: "Comparison of observed continental- and global-scale changes in surface temperature with results simulated by climate models using natural and anthropogenic forcings. Decadal averages of observations are shown for the period 1906 to 2005 (black line) plotted against the centre of the decade and relative to the corresponding average for 1901–1950. Lines are dashed where spatial coverage is less than

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50%. [Dark] shaded bands show the 5–95% range for 19 simulations from five climate models using only the natural forcings due to solar activity and volcanoes. [Light] shaded bands show the 5–95% range for 58 simulations from 14 climate models using both natural and anthropogenic forcings." *Source*: *Climate Change 2007: The Physical Science Basis* (Cambridge University Press, 2007). Image courtesy of IPCC. Colors replaced as noted in brackets to allow non-color reproduction.

Do the results shown in figures 13.2 and 13.3 "verify" or "validate" climate models? No, because Oreskes et al.'s critique of those terms still applies. Under that analysis, it is conceivable (though unlikely) that somehow all the various modeling groups have independently parameterized and/or tuned their models to bring them into close convergence with the twentieth century trend. If not, it is still conceivable (though again unlikely) that for reasons we cannot anticipate, it will turn out that none of these independently parameterized models predict the climate accurately. Ultimately, neither comparing climate models with observations nor comparing them with each other can prove that they are correct. Nor would any climate modeler ever make such a claim: models are, after all, simulations.

Yet as a principled and detailed mode of evaluation, the comparisons can help us *trust* the models, even if they do not provide proof in any strong sense. Here the epistemological undercurrents of this book's argument have profound practical implications. Distinguishing evaluation and confirmation from validation or verification helps to clarify the proper role of models in forecasting climatic change: not as absolute truth claims or predictions, but as heuristically valuable simulations or projections.[27]

The picture that I hope is emerging here is that *all* knowledge about climate change depends fundamentally on modeling. It's not that there is no such thing as an observation separate from modeling. It's that putting together a trustworthy and detailed data image of the global climate—getting enough observations, over a long enough time span—requires you to model the data, to *make* them global. It's not that climate simulation models are perfectly reliable, any more than weather forecast models always get it right. Instead, it's that the simulations already include a lot of data in their parameters; they are precisely *not* pure theories, but theories already partially adjusted to the conditions we observe in the real world. That's model-data symbiosis. So whether we are looking at data images or model projections of climate futures, we will always experience them as probabilistic, as shimmering rather than fixed.

At the end of chapter 10, I talked about ensemble forecasting—the deliberate use of multiple, slightly perturbed data analyses to create a set of forecasts that then can be studied as a statistical ensemble. The point there was that once you understand that no observations and no data analysis will ever be a perfect image of the real atmosphere, you can stop trying to make them so. Instead, you can exploit the certainty of imperfection by making your forecast from many slightly different data images: shimmering data.

Complicating this picture even further, in terms relevant to parameterization, is the most recent development in ensemble prediction: perturbed model physics, now being explored by both weather and climate modelers. This is the logical counterpart to perturbed data analysis. Its premise is that models are at least as likely as observations to contain errors, particularly in parameterized quantities. Often the physically plausible range of a parameter is well established, but not its exact value. To exploit this fact, perturbed model physics systematically varies the values of important parameters within their known ranges. In complex models with many parameters, the number of possible combinations of parameter values is large; each combination constitutes a variation of the

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model. The perturbed-physics technique thus generates an ensemble that can include dozens to thousands of model variations (or more). It then runs each variation and compares the outputs of all the runs statistically. In effect, this procedure replaces subjective tuning with systematic variation of parameter values.

One project developing this technique, climate*prediction*.net, began around 2000 at Oxford University and other institutions in the United Kingdom, including the UK Met Office and the University of Reading. Climate*prediction*.net links thousands of personal computers into a gigantic "grid computer," functionally equivalent to a supercomputer, over the Internet. At this writing, its computing grid claims more than 100,000 individual host computers, whose owners volunteer idle computer time for climateprediction.net to run its climate models. The project has already made tens of thousands of runs of various Hadley Centre models on a variety of modeling experiments using perturbed physics, and has begun to issue publications.

Originally climate*prediction*.net hoped to narrow the predicted range of climate sensitivity by using the perturbed-physics technique. Instead, initial results *widened* the range rather dramatically. An early study using more than 2000 runs produced a climate sensitivity range of 1.9°C to *11.5°C* (an upper bound almost 2½ times higher than that of the IPCC).[28] However, a later, better-constrained study from the same group reduced that range to 2.2–6.8°C.[29]

More than 2000 variations on the same model, 2000 possible versions of atmospheric physics, 2000 potential planetary futures—what do these shimmering images tell us? It is too early to evaluate the perturbed-physics technique definitively. For one thing, despite the thousands of variations, these runs used only one of the available GCMs; perturbed-physics runs on other GCMs might reach different outcomes. Still, the relatively greater agreement among climate models used in the IPCC reports could conceivably be due to questionable parameterization and tuning practices—"bad empiricism with a bag over its head," as Randall and Wielicki put it. Yet even if true, this argument cuts against climate-change skepticism. Most skeptical arguments implicitly or explicitly promote the idea that warming is minimal or nonexistent and that greenhouse warming is a false threat. Yet *not one* of the perturbed-physics model runs predicted warming of less than 1.9° C. No combination of parameters produced anything like a flat or downward trend. Instead, the main effect of these runs was to raise the *upper* end of the sensitivity range. In other words, if these runs show anything, they show that the IPCC sensitivity estimate—currently 2–4.5°C—may be much too conservative.

Like data about the climate's past, model predictions of its future shimmer. Climate knowledge is probabilistic. You will never get a single definitive picture, either of exactly how much the climate has already changed or of how much it will change in the future. What you will get, instead, is a range. What the range tells you is that "no change at all" is simply not in the cards, and that something closer to the high end of the range—a climate catastrophe—looks all the more likely as time goes on.

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